INCH-POUND

MIL-M-80226B
1 August 1990
SUPERSEDING
MIL-M-80226A
26 November 1984

MILITARY SPECIFICATION

METALLIZING SYSTEM, ELECTRIC ARC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers metallizing systems which apply metallic coatings in the form of molten spray. The molten spray is developed by an electric arc generated between the ends of two electrically charged wires. High velocity compressed air or other suitable gases atomizes the molten metal and projects it onto the workpieces to be coated.

1.2 Classification. This specification covers metallizing systems of one type and the following sizes. The size to be furnished shall be as specified (see 3.7 and 6.2).

Size 1A - 200 amperes at 20 to 38 volts.
Size 2A - 350 amperes at 20 to 38 volts.
Size 3 - 500 amperes at 20 to 40 volts.
Size 4 - 650 amperes at 20 to 48 volts.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd. Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or letter.

AMSC N/A FSC 3433
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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated copies of Federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5054.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U.S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

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(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contract activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AEROSPACE MATERIAL SPECIFICATIONS (AMS)

AMS 4026 - Aluminum Alloy Sheet and Plate.

AMS 5504 - Steel Sheet, Strip, and Plate, Corrosion and Moderate Heat Resistant (DOD Adopted).

(Application for copies should be addressed to Customer Services Department, Publications Group, Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale PA, 15086-7511.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM C 633 - Adhesion or Cohesive Strength of Flame Sprayed Coating (DOD Adopted).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery (DOD Adopted).

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete metallizing system shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

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3.2 Design. The metallizing system shall be new (not a prototype) and one of the manufacturer's current production models capable of spraying molten metals developed in an arc by simultaneously atomizing two electrically charged wires. The system shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair. The system shall be designed and constructed so that it may be shipped fully assembled and ready for use when connected to the proper electric and pneumatic services.

3.2.1 Measuring and indicating device calibration. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.2 Reclaimed materials. The metallizing system may contain reclaimed materials to the extent possible provided such materials will not jeopardize the intended use, performance, or design life of the system. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The metallizing system and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Safety and health requirements. The metallizing system shall be equipped with safety devices as required to protect operating personnel from electrical voltages constituting a safety hazard. Ground terminals shall be provided on the chassis of the control console and power supply. Power supply terminals shall be enclosed in a manner that will prevent accidental contact by operating personnel. The entire shell of the spray gun assembly shall be electrically neutral or insulated to insure that the operator will not be exposed to electrically charged components. An automatic means shall be provided to prevent application of electric power to the spray gun when air pressure is less than required for safe operation of the system as specified by the manufacturer. Equipment safety devices shall include appropriate devices to protect the direct current (DC) power supply from overload. All parts, components, mechanisms, and assemblies furnished on the metallizing system, whether or not specifically required herein, shall conform with all requirements of 29 CFR 1910. Additional safety and health requirements shall be as specified in detail (see 6.2). All safety devices normally furnished as standard on the manufacturer's commercial metallizing system supplied to the commercial market shall be provided.

3.2.5 Mercury restriction. The metallizing system shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

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3.2.6 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the metallizing system.

3.2.7 Environmental protection. During the manufacture of the metallizing system and its components, materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect on the date of the contract shall not be used or emitted.

3.2.8 Lubrication. Means shall be provided to ensure lubrication for all moving parts.

3.2.9 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine. All replacement parts shall be interchangeable with their respective original parts, and fit/function with mating components.

3.3 Construction. The metallizing system shall be constructed of parts which are new, without defects, and free of repairs.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the metallizing system shall be painted in accordance with the manufacturer's commercial practice and color.

3.3.6 Threads. All threaded parts used on the metallizing system and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

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3.4 Components. The metallizing system shall consist of not less than the following components.

3.4.1 Gun. Unless otherwise specified (see 6.2), the spray gun shall be designed to spray wire 0.063 + 0.002 inch in diameter. Wire guides that also serve as electrodes shall be provided to guide the wires to the converging point where the arc is established. Electric power to the wires shall be provided through the electrodes. The design of the electrodes shall provide for minimum wear and heat abrasion, and for maximum transfer of electric power to the wires. The compressed air atomizing nozzle shall be positioned for maximum concentration and efficiency of the spray pattern. The gun shall be equipped with all fittings and connections required for feeding the wire into the gun through the wire guides. All necessary connections shall be provided for connection of electrical power, compressed air or gas, and control circuits. A gun mounted control or controls shall start and stop air or gas, wire feed, and arc current. Unless otherwise specified (see 6.2), the gun shall be designed for machine mounting or hand use. The maximum weight of the gun, not containing wire drive units shall not exceed 5 pounds. The weight of the gun, containing built-in wire drive units shall not exceed 10 pounds. The above weights are exclusive of electrical cables, wire feed conduits, and air hose.

3.4.2 Wire feed mechanism. Unless otherwise specified (see 6.2), the system shall include a dual wire feed mechanism which is an integral part of the spray gun. The wire mechanism shall be powered by an infinitely variable speed electric motor, air motor, or air turbine. The motor or turbine shall have adequate power to feed the wire at a constant rate from fully loaded reels at any speed within the range of the feed unit. Wire feed rates shall be provided to meet the deposition rates listed in table I. The drive roll pressure shall be adjustable to assure positive wire feed. The drive roll shall be insulated from the gun body or case.

TABLE I. Deposition rate at continuous duty.

Size	Pounds per hour (minimum)				
	Aluminum	Stainless steel	Zinc	Steel	Ni 95/AL 5
1A	11	16	22	18	20
2A	21	35	84	35	35
3	25	48	115	50	50
4	30	60	140	65	65

3.4.3 Wire reels and stand. The wire feed mechanism shall have two separate reels for holding coils of wire. Each reel shall be mounted on a stand and readily accessible for easy loading. The reels shall be electrically insulated from the stand and from each other. A friction brake on each reel shall control the tension of the wire drag. The stand shall be mounted on casters. Each reel shall be capable of holding and operating with a wire coil weight of not less than 25 pounds.

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3.4.3.1 Wire straightners. When specified (see 6.2), wire straightners shall be furnished. The straightners shall be capable of removing the set curvature from any coiled or spooled wire used with the system.

3.4.3.2 Wire feed controls. The wire feed controls shall be located in a convenient and accessible position on the gun or control panel. The wire feed rate shall be adjustable manually to any speed within the operating range of the feed system. The controls shall provide fine adjustment of wire feed and shall include an instant "stop-start" mechanism that does not change the feed rate. The system shall prevent fusing of the wires when spraying.

3.4.4 Power supply (DC). The power supply shall be designed for operation with electric arc wire metallizing equipment and shall be regulated as necessary to provide DC power having the required constant voltage characteristics. The rated continuous duty DC output current (100 percent duty cycle) and voltage shall be as specified in table II. The output voltage shall be continuously variable throughout the range of the power supply. The power supply shall be either wheel or skid mounted. The power supply shall be enclosed in a cabinet of steel frame and panel construction. The gage of the metal, and the physical strength of the chassis and cabinet shall be adequate for support and protection of all components. The side and top of the cabinet shall be removable to provide access for servicing. When forced air cooling is provided, the fan motor shall be designed to operate from the same electric power source as the power supply. The system shall be capable of operating under full load in an ambient temperature of 104 degrees Fahrenheit (F).

TABLE II. Power supply output.

Size	Amperes 1/	Range 2/
1A	200	20 to 38
2A	350	20 to 38
3	500	25 to 40
4	650	20 to 48

1/ The ampere rating of the power supply shall meet or exceed the value stated.
2/ Where a range is shown, the voltage available may be less than the lowest voltage stated or more than the maximum voltage stated but must be capable of performing within the stated range.

3.4.4.1 Power supply controls and indicators. The power supply controls shall include an "on-off" or starting switch and an output voltage control. A DC Voltmeter and ammeter for measuring the output voltage and current shall be included and mounted on the control panel or power supply. The accuracy of the meters shall be not less than ± 3 percent of full scale.

3.4.5 Air filtration and control. The metallizing system shall be equipped with an air filtration and control system. The air filtration system shall include a cleanable or replaceable air filter and an oil-water separator. The air control system shall be equipped with not less than a control valve,

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pressure regulator, air pressure gauges for atomizing air, and when applicable, controls for the wire feed air motor or air turbine. The air filtration and control equipment may be located either in a separate unit or be an integral part of the wire control unit or power supply.

3.4.6 Hose and power cables. All necessary hoses and power cables for interconnecting the metallizing system shall be furnished. The hoses and power cables shall be not less than 15 feet in length. Insulated flexible conduit shall be furnished to encase the wires between reel stands or straighteners and the spray unit. The flexible conduit shall be not less than 7-1/2 feet in length.

3.4.7 Hour meter. When specified (see 6.2), the metallizing system shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the power supply. The meter shall be of the nonresetting type and shall have a range of 0 to 9,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture.

3.5 Electrical requirements. The machine shall be wired to conform to NFPA 79. Unless otherwise specified (see 6.2), each machine shall be designed to operate on 3-phase, 60-Hertz (Hz), 230/460-volts. Unless otherwise specified (see 6.2), the metallizing system shall be initially wired for operation on 460 volts.

3.6 Air or gas supply. The metallizing system shall operate at maximum rated capacity from an air source which shall not exceed 60 cubic feet per minute (CFM) and a maximum pressure of 110 pounds per square inch (PSI).

3.7 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the metallizing system shall meet the requirements of table I and table II for the size specified. The metallizing system size and capacity shall be not less than the stated requirements of the size ordered.

3.8 Performance. The metallizing system shall be capable of melting and spraying in wire form the materials listed in table I. The deposition rates shall be not less than specified in table I.

3.8.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.9 Standard equipment. All standard equipment normally provided with the manufacturer's commercial metallizing system shall be furnished.

3.10 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2). The optional equipment provided shall be functional without requiring modification of the metallizing system.

3.11 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

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3.12 Marking on instruments, control panels, charts, and plates. All words on the metallizing system including all instruments, control panels, control units, charts, plates, and related attachments shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant

3.12.2 Nameplate. A nameplate shall be securely attached to each metallizing system. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract number or order number
National stock number or plant equipment code
Date of manufacture

3.13 Workmanship. Workmanship of the metallizing system and its accessories shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

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4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.7.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. The metallizing system shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each metalizing system prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.2, and the inspection in 4.8. The machine shall pass the examination, the test, and the inspection to be accepted.

4.6 Examination. Each metallizing system shall be examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41^0 to 104^0 Fahrenheit (F) and 20 percent to 95 percent relative humidity.

4.7.2 Operational test. The metallizing system shall be connected to electric and air sources. The system shall then be prepared for operation in accordance with the manufacturers published operating instructions. The system shall be tested under operating conditions for not less than 30 minutes. Performance of all components which make up the system shall be checked for proper operation. During the test, a test specimen conforming to AMS 4026 (aluminum) and one specimen conforming to AMS 5504 (steel) shall be sprayed with a wire selected from table I. The sprayed coatings shall have a uniform appearance. Surface defects of the coatings shall be limited to small nodules not to exceed 0.045 inch in diameter and shall not exceed 0.015 inch above the surrounding sprayed surface. The coatings shall not contain any:

Blisters
Cracks
Chips or loosely adhering particles
Pits exposing the substrata
Any other evidence of malfunction

Any evidence of faulty or erratic operation shall be cause for rejection.

4.7.3 Performance tests. Unless otherwise specified (see 6.2), the system shall be tested for metallizing capabilities by spraying two test specimens conforming to AMS 4026 and two specimens conforming to AMS 5504. Each specimen shall be sprayed with a different wire selected from table I as specified by the procuring activity (see 6.2). Unless otherwise specified (see 6.2) preparation of the specimens and the spray technique shall be the responsibility of the manufacturer and as described in their operating

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manuals. The system to be tested shall be prepared as specified in 4.7.2. The sprayed coatings shall conform to 4.7.2 Sufficient spraying shall be accomplished to verify that the spray rate for each material is not less than specified in table I.

4.7.3.1 Bond strength test. When specified (see 6.2), a bond strength test shall be accomplished in accordance with ASTM C 633. The substrate sprayed, wire sprayed, and test limits shall be as specified by the purchaser (see 6.2).

4.7.3.2 Ammeter and voltmeter test. The ammeter and voltmeter shall be tested to determine conformance with 3.4.4.1. A certificate of calibration issued by the manufacturer may be submitted in lieu of this test.

4.7.3.3 Additional performance tests. Performance tests required in addition to or in lieu of the requirements of 4.7.3 shall be as specified and fully described (see 6.2).

4.7.4 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.7 using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirement for packaging shall be in accordance with MIL-W-45562.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The metallizing systems described herein are intended for use in facilities having a requirement for medium or high speed metal spraying capability.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. Specify system of units for measuring and indicating devices (see 3.2.1).

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f. Additional safety and health requirements, if required (see 3.2.4).

g. Painting, if different (see 3.3.5.).

h. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.7).

i. Specify wire sizes, if different (see 3.4.1).

j. Specify type of mounting, if different (see 3.4.1).

k. Specify type of wire feed, if different (see 3.4.2).

l. Specify wire straightner, if required (see 3.4.3.1)

m. Hour meter, if required (see 3.4.7).

n. Electrical requirements, if different (see 3.5).

o. Voltage requirements, if different (see 3.5).

p. Size and capacity, if different (see 3.7).

q. Additional performance features, if required (see 3.8.1).

r. Specify and fully describe optional equipment, if required (see 3.10).

s. Specify applicable repair parts, if required (see 3.11).

t. Nameplate, if different (see 3.12.2).

u. Inspection conditions, if different (see 4.3).

v. First article inspection, if different (see 4.4).

w. Quality conformance inspection, if different (4.5).

x. Performance test, if different (see 4.7.3).

y. Type wire to be used in performance test (see 4.7.3).

z. Specific instructions for preparation of test samples and spray technique, if different (see 4.7.3

aa. Bond strength test and parameters of test, if required (see 4.7.3.1).

bb. If bond strength test is required, specify substrate, wire, and test limits (see 4.7.3.1).

cc. Additional performance tests, if required (see 4.7.3.3).

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6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specification acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
N/A	DI-TMSS-80527	Commercial Off-The-Shelf (COTS) Manuals	Require all data be in The English Language
N/A	DI-TMSS-80528	Supplemental Data for Commercial Off-The-Shelf (COTS) Manuals	Require all data be in The English Language

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Technical data. It should be specified in the remarks section of the DD Form 1423 that all technical data furnished should be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International d'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

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6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Cross-reference of classification. Classification changes from the previous issue of this specification are as follows:

MIL-M-80266	MIL-M-80226A	MIL-M-80266B
One size	Size 1 (new size)	Deleted
	Size 2 (new size)	Deleted
	Size 3 (new size)	Size 1A (new size)
		Size 2A (new size)
		Size 3
		Size 4 (new size)

6.8 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

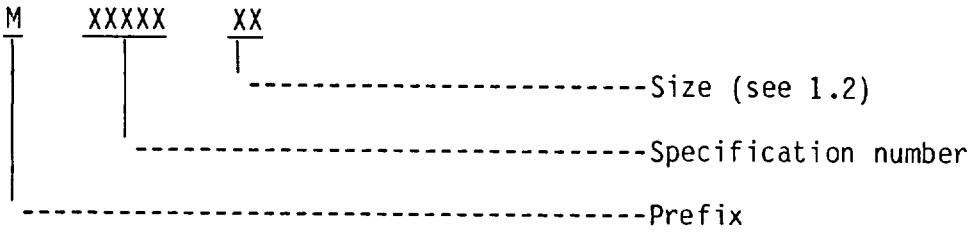
6.9 Safety and health requirements. 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.10 Training. Training required by the procuring activity should be provided as specified in the contract.

6.11 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.12 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.13 Part or Identifying Number (PIN). The PIN to be used for metallizing systems acquired to this specification are created as follows:



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6.14 Subject term (key word) listing.

Aluminizing
Coating
Metal protection
Metal spraying
Plating
Spray coating
Substratas
Surface finishing

Custodians:
Army - ME
Navy - SH
Air Force - 99

Review activities:
Air Force - 84
DLA - GS

User:
Navy - MC

Preparing activity:
DLA - IP

Project (3433-0117)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-M-80226B	90 Aug 01

3. DOCUMENT TITLE
Metallizing System, Electric Arc

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)*	
James D. Threadgill/DIPEC-SSG	(1) Commercial (901)775-4749	(2) AUTOVON 683-4749
c. ADDRESS *(Include Zip Code)* Defense Industrial Plant Equipment Ctr 2163 Airways Blvd. Memphis, TN 38114-5051	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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